

last class: Induction

Induction: sequence of mathematical statements

strong induction:

Consider a sequence of mathematical statements

$(s_1, s_2, s_3, \dots)$

- suppose s_1 is true
- suppose for $k \in \mathbb{N}$ if $s_1, s_2, \dots, s_k$ all are true
Then s_{k+1} is true,

Then, s_n is true $\forall n \in \mathbb{N}$

Key difference:

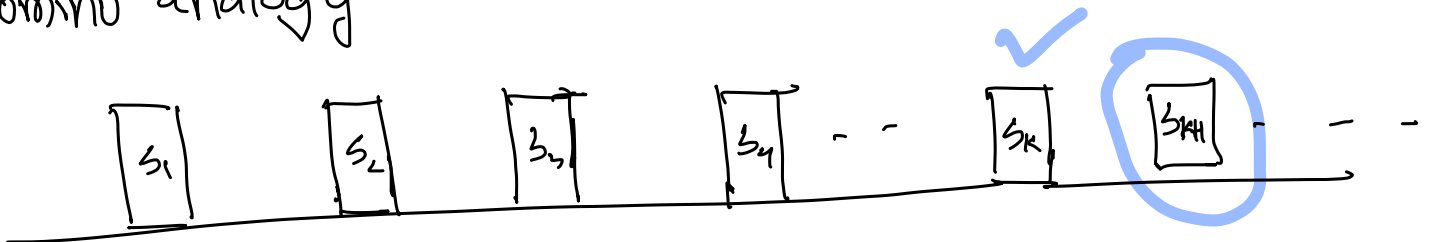
Induction:

suppose for $k \in \mathbb{N}$ if s_k is true then s_{k+1} is true

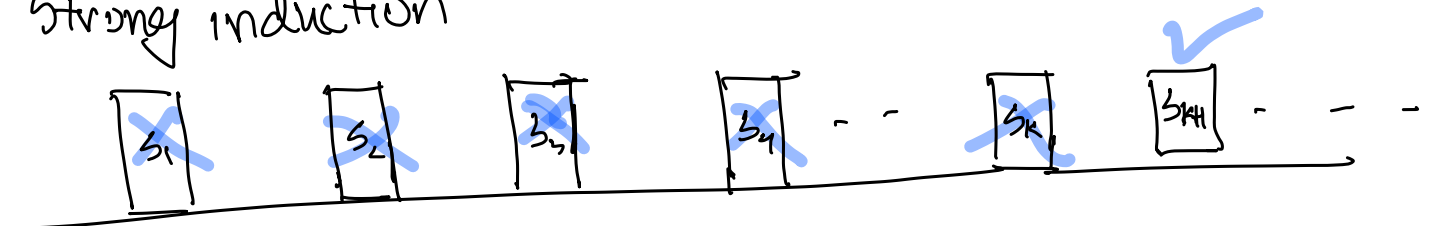
strong induction:

suppose for $k \in \mathbb{N}$, $s_1, s_2, \dots, s_k$ is true then s_{k+1} is true

Domino analogy



strong induction



Proof structure:

Proposition: $s_1, s_2, s_3, \dots$ are all true

Proof:

- Base case: show, s_1 is true
- Inductive hypothesis: For $k \in \mathbb{N}$, we assume,
 $s_1, s_2, \dots, s_k$ all are true
- Induction setup:
if $s_1, s_2, \dots, s_k$ all are true
then s_{k+1} is true
- conclusion:
Therefore by induction all the s_n are true

Theorem: (Fundamental theorem of arithmetic)

Every integer $n \geq 2$ is either a prime or product of primes

Proof:

Base case: $n=2$, since 2 is a prime so,
we are done (s_1 is true)

Inductive hypothesis:

we consider, $k \in \mathbb{N}$ and $k \geq 2$, we assume
that each integer $2, 3, 4, \dots, k$ are
either prime or product of primes,

Induction step:

we need to show $(k+1)$ is prime or product of primes given that, $2, 3, \dots, k$ are primes or product of primes

Why strong induction not normal induction:

normal induction: we assume k is prime or product of primes

tells us nothing about whether $k+1$ is prime or product of primes

rather it gives us information regarding $2k, 3k, 4k, \dots$ are product of primes

$k+1$ is greater than 1 so it can either be prime or a composite number,

Case 1: $k+1$ is prime, [ϕ_{k+1} is true]

Case 2: $k+1$ is composite,

Let's say, $k+1 = s \cdot t$

where,

$1 < \underline{s} < k+1$ & $1 < \underline{t} < k+1$

s, t : prime or product of primes

(if we proceeded with normal induction we wouldn't have made any progress with s and t)

$$\therefore s = p_1^{l_1} p_2^{l_2} \dots p_n^{l_n}$$

$$t = q_1^{k_1} q_2^{k_2} \dots q_m^{k_m}$$

$$\text{primes} \begin{cases} p_1, p_2, \dots, p_n < s \\ q_1, q_2, \dots, q_m < t \end{cases}$$

- $l_1, l_2, \dots, l_n$: non negative integers
- $k_1, k_2, \dots, k_m$: non negative integers

$$\therefore k+1 = s \cdot t$$

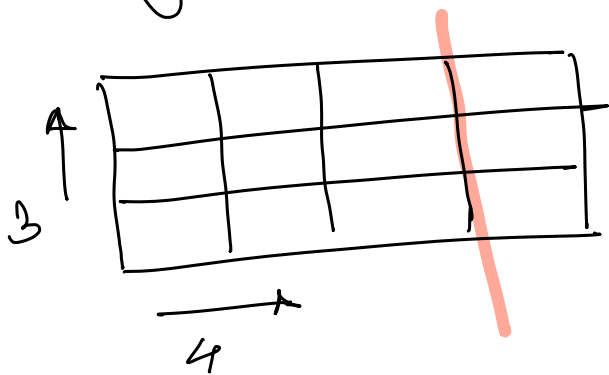
$$= (p_1^{l_1} p_2^{l_2} \dots p_n^{l_n}) \cdot (q_1^{k_1} q_2^{k_2} \dots q_m^{k_m})$$

$\therefore k+1$ is a product of primes

hence (s_{k+1}) statement is true

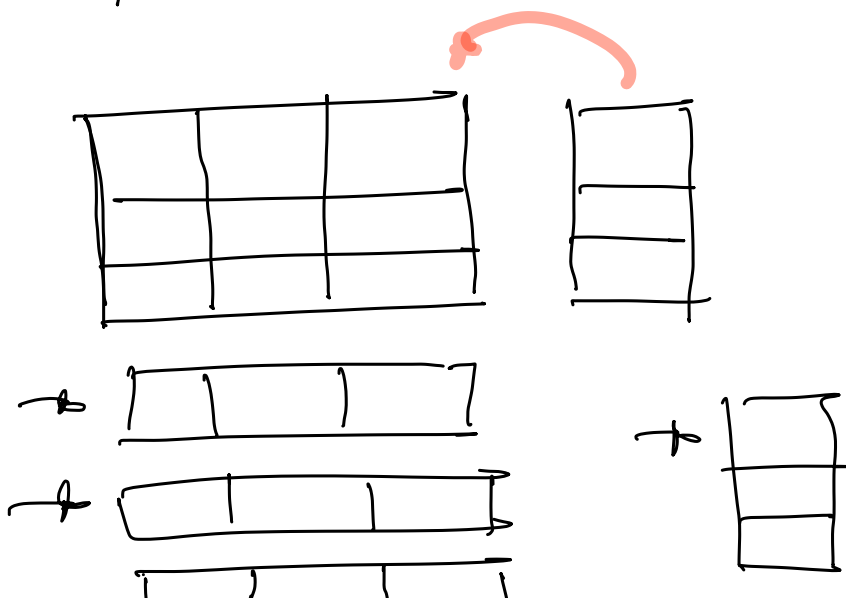
Conclusion: s_n is true for $\forall n \in \mathbb{N}$

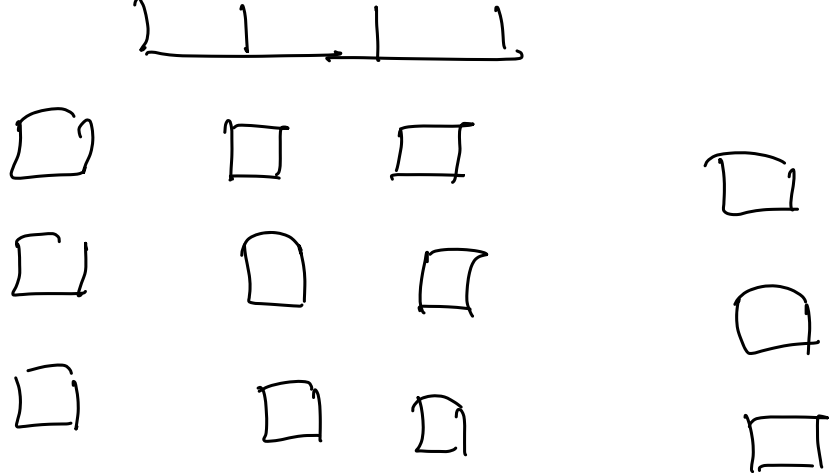
Breaking of a chocolate bar (Jay Cummings)



4x3 grid bar

$$4 \times 3 = 12$$

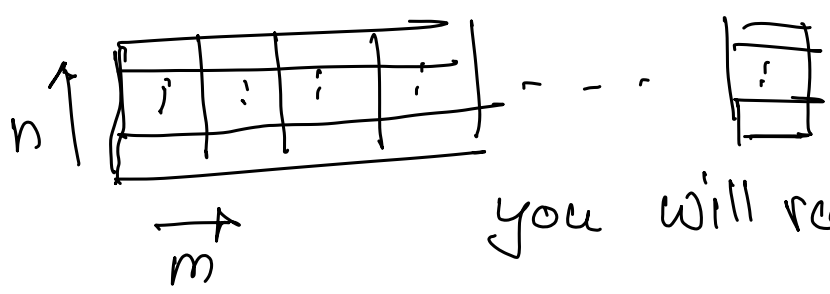




Total: 11

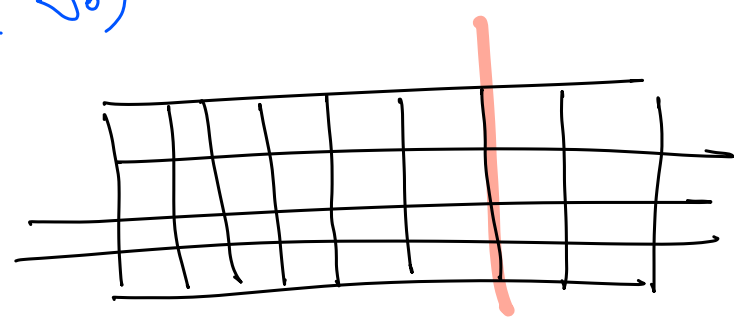
$$(12 - 1) = 11$$

in general:



you will require $m \cdot n - 1$ steps to break down into single chunks

strong induction
(why?)



base case, 1 chunk



$$1 - 1 = 0$$

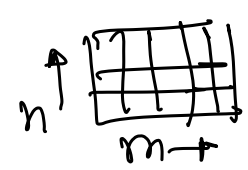
inductive hypothesis:

(normal)

we assume $m \cdot n$ chunks

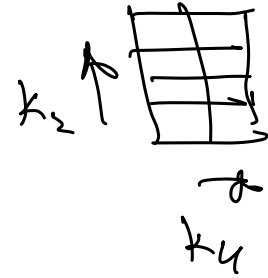
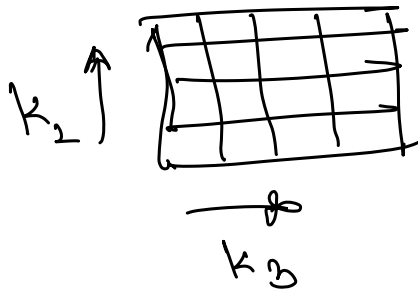
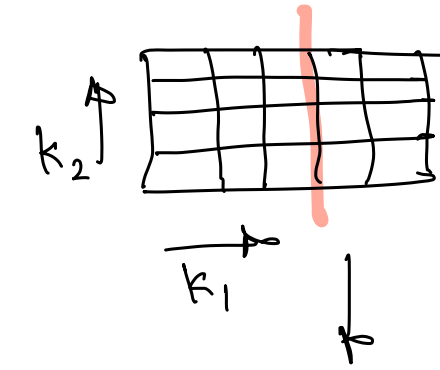
we require,

$m \cdot n - 1$ steps



induction $m \cdot n + 1$ chunk of chocolate
step 2

$k_1 \cdot k_2 = m \cdot n + 1$



strong induction :

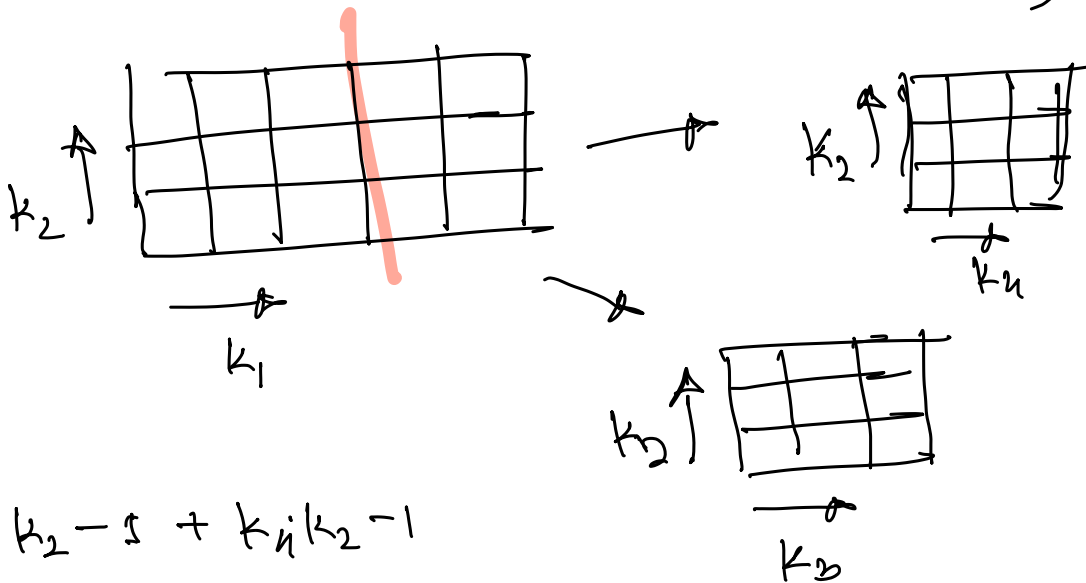
grid
size

1, 2, 3, ..., $m \cdot n$ all can be

decomposed into
individual chunks
by $m \cdot n - 1$ steps

no chunk
of chocolates:

Now In induction step ($m \cdot n + 1$ case)



$\frac{k_4 \cdot k_2 \text{ grid}}{\hookrightarrow}$
 $k_4 \cdot k_2 - 1$
steps

$\frac{k_3 \cdot k_2 \text{ grid}}{\hookrightarrow}$
 $k_3 \cdot k_2 - 1$ steps

$$k_3 \cdot k_2 - 1 + k_4 \cdot k_2 - 1$$

$$= k_2 (k_3 + k_4) - 1 - 1$$

$$= \boxed{k_1 \cdot k_2 - 1 - 1}$$

$$= m \cdot n + 1 - 1 - 1 = \boxed{m \cdot n - 1}$$

Multiple base cases:

Base case: $n=1, \textcircled{2}, \dots$
we only required 1

sometimes, base case, $n=1$ becomes to trivial

strong induction, $s_1, s_2, s_3, \dots, s_k \rightarrow s_{k+1}$
base case

n	strong induction (typical)	strong induction 2 step	After this step we know
1	base case	base case	s_1 is true
2	$s_1 \rightarrow s_2$	<u>base case</u>	$s_1, s_2 : \top$
3	$s_1, s_2 \rightarrow s_3$	$s_1, s_2 \rightarrow s_3$	$s_1, s_2, s_3 : \top$
4	$s_1, s_2, s_3 \rightarrow s_4$	$s_1, s_2, s_3 \rightarrow s_4$	$s_1, \dots, s_4 : \top$
	$\vdots$		

Proposition: Every $n \in \mathbb{N}$ with $n \geq 1$ can be written as
 $2a + 5b$ for some natural numbers a and b

Sketch of proof:

$$11 = 2 \cdot 3 + 5 \cdot 1 \quad 13 = 2 \cdot 4 + 5 \cdot 1 \quad 15 = 2 \cdot 5 + 5 \cdot 1$$

$$12 = 2 \cdot 1 + 5 \cdot 2 \quad 14 = 2 \cdot 2 + 5 \cdot 2 \quad 16 = 2 \cdot 3 + 5 \cdot 2$$

Proof:

Base case:

$$11 = 2 \cdot 3 + 5 \cdot 1$$

$$12 = 2 \cdot 1 + 5 \cdot 2$$

for the statement For both of the base cases
is true

Inductive hypothesis:

We assume for, $k \in \mathbb{N}$, $k \geq 11$

$11, \dots, k$

can be expressed as $2a + 5b$, $a, b \in \mathbb{N}$

Induction step:

$11, 12, \dots, k$ can be expressed as $2a + 5b$, $a, b \in \mathbb{N}$

then, we have to show, $k+1$ can also
be expressed as $2a + 5b$, $a, b \in \mathbb{N}$

hence, $k-1 < k$

$$\therefore k-1 = \underline{2a + 5b} \quad [a, b \in \mathbb{N}]$$

Hence

$$k+1 = \underline{k-1} + 2 = \underline{2a} + 5b + \underline{2} = 2(a+1) + 5b$$

$$\because a \in \mathbb{N}, \quad a+1 \in \mathbb{N}$$

$\therefore k+1$ can be expressed as $2a + 5b$, $a, b \in \mathbb{N}$

Conclusion:

b_n is true, $\forall n \in \mathbb{N}$

Theorem:

Every $n \in \mathbb{N}$ can be expressed as a sum of distinct powers of 2 in precisely one way,

Proof:

Base case: $n=1$ the statement holds since

$$1 = 2^0$$

and this is the only to express 1 as a sum of powers of 2

all the higher powers of 2 are greater than 1

Inductive hypothesis:

for, 1, 2, ... k can be expressed as sum of powers of 2 in only one way.

Induction step:

we need to show $k+1$ can be expressed as sum of powers of 2 in only one way

Case 1: $k+1 = 2^m$

this is the only way, to express $k+1$ as sum of powers of 2

since, $k+1 = 2^m \Rightarrow k = 2^m - 1$

from last class we learnt,

$$1 + 2 + 4 + 8 + \dots + 2^n = 2^{n+1} - 1 \quad (\text{Lemma})$$

I can write,

$$k = 1 + 2 + \dots + 2^{m-1}$$

then m

by using smaller non negative integers we can at maximum construct k, let alone k+1

Case 2

if $k = 2^m$, $k+1 = 2^m + 2^0$

since by inductive hypothesis this is the only way to construct k

so, this is the only way to construct k+1

Case 3:

if, $2^m < k$, we will consider, m: choice

$(k+1) - 2^m$, which is less than 2^m

↓

if this statement were not true

$$2^m + 2^m \leq (k+1)$$

$$= 2 \cdot 2^m \leq (k+1)$$

$$= 2^{m+1} \leq (k+1)$$

choice of m was such that, $2^m < k < 2^{m+1}$

hence, $(k+1) - 2^m < 2^m < k$

by induction hypothesis,

$(k+1) - 2^m < k$ it can be expressed as a sum of power 2 in a single way

$$\underline{(k+1) - 2^m} < k$$

expressed as sum of powers of 2 in a single way,

$$\therefore (k+1) - \underline{2^m} = 2^{m_1} + 2^{m_2} + \dots + 2^{m_n}$$

$$\rightarrow \underline{(k+1)} = \underline{2^m} + 2^{m_1} + 2^{m_2} + \dots + 2^{m_n}$$

$\therefore$ (s_{k+1} statement)

Conclusion:

b_n : true for, $n \in \mathbb{N}$

Theorem: Binomial theorem:

For, $x, y \in \mathbb{R}$ and $n \in \mathbb{N}_0$

$$[\mathbb{N}_0 = \{0\} \cup \mathbb{N}]$$

$$(x+y)^n = \sum_{m=0}^{m=n} \binom{n}{m} x^{n-m} y^m$$

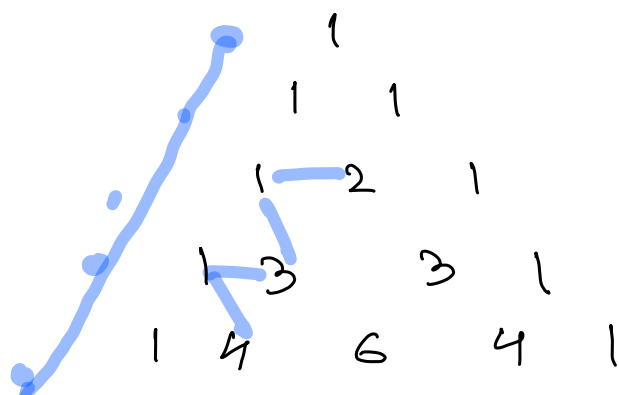
where $\binom{n}{m}$: binomial coefficient defined as $\frac{n!}{m!(n-m)!}$

Proof:

Base case: $n=0$, $(x+y)^0 = 1$

$$n=1, (x+y)^1 = x+y$$

in order
for this
tree
to work



but
we need
strong
induction k^{th}

$$\binom{k}{0} \quad \binom{k}{1} \quad \dots \quad \binom{k}{r-1} \quad \binom{k}{r} \quad \dots \quad \binom{k}{k}$$

$$\checkmark \quad k+1^{th} \quad \binom{k+1}{0} \quad \binom{k+1}{1} \quad \dots \quad \binom{k+1}{r} \quad \dots \quad \binom{k+1}{k+1}$$

$$\boxed{\binom{k+1}{r} = \binom{k}{r-1} + \binom{k}{r}}$$

inductive hypothesis:

$0, 1, 2, \dots, k$ we assume the
formula holds,

$$\therefore (x+y)^k = \binom{k}{0} x^k + \binom{k}{1} x^{k-1} y^1 + \dots + \binom{k}{r} x^{k-r} y^r + \dots + \binom{k}{k} x^{k-k} y^k$$

induction step: For, $k+1$

$$(x+y)^{k+1} = (x+y) \cdot (x+y)^k$$

$$= (x+y) \left\{ \binom{k}{0} x^k + \binom{k}{1} x^{k-1} y^1 + \binom{k}{2} x^{k-2} y^2 + \dots + \binom{k}{k} x^{k-k} y^k \right\}$$

$$= \binom{k}{0} x^{k+1} + \binom{k}{1} x^k y^1 + \binom{k}{2} x^{k-1} y^2 + \dots + \binom{k}{k} x y^k + \binom{k}{0} x^k y^1 + \binom{k}{1} x^{k-1} y^2 + \dots + \binom{k}{k} y^{k+1}$$

$$= \binom{k}{0} x^{k+1} + \left\{ \binom{k}{0} + \binom{k}{1} \right\} x^k y^1 + \left\{ \binom{k}{1} + \binom{k}{2} \right\} x^{k-1} y^2 + \dots + \binom{k}{k} y^{k+1}$$

$$= \binom{k+1}{0} x^{k+1} + \binom{k+1}{1} x^{k+1-1} y^1 + \binom{k+1}{2} x^{k+1-2} y^2 + \dots + \binom{k+1}{k+1} y^{k+1}$$

$$= \sum_{m=0}^{k+1} \binom{k+1}{m} x^{k+1-m} y^m$$

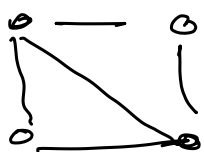
$\therefore (b_{k+1} \text{ is } T)$

Conclusion: $b_n, \text{ True } b_n \forall n \in \mathbb{N}$

Ferdinand's Rittle theorem

Kentel's theorem

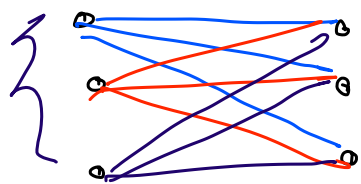
a graph with vertex at maximum we can have n^2 edges without forming a triangle
 n^2+1 edges will always guarantee there is a triangle



4 : vertex $(2 \cdot 2)$

$$2^2 = 4 \quad 4+1$$

$2n$ vertex



bipartite
graph

$$3 \cdot 3 = 9$$

still no triangle

(2n) even of vertex in a graph

↳ bipartite graph

maximally connect without
forming triangle

n^2+1 will always guarantee me
with a triangle,

Theorem: Mantel's theorem

if a graph G has $2n$ vertices and n^2+1 edges
then G contains a triangle

Proof:

base case: $n=1$, $2 \cdot 1 = 2$: vertex

$1^2+1 = 2$: edges



there is no graph with 2 vertex and 2 edges

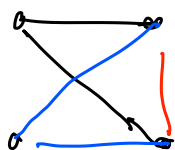
so the statement vacuously holds,

$(F \Rightarrow T)$
 T

$$n=2,$$

$$2 \cdot 2 = 4 \text{ vertex}$$

$$2^r + 1 = 5 \text{ edges}$$



there is a triangle,

Inductive hypothesis:

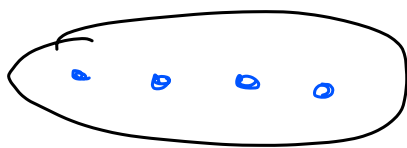
For, $k \in \mathbb{N}$ we assume
every graph on $2k$ vertices and
 k^2 edges
contains a triangle,

Induction step:

We now prove, a graph G $2(k+1) = (2k+2)$
vertices

and, $(k+1)^2 + 1$ edges contains a triangle

$2k$, 2 u v
 2 vertices

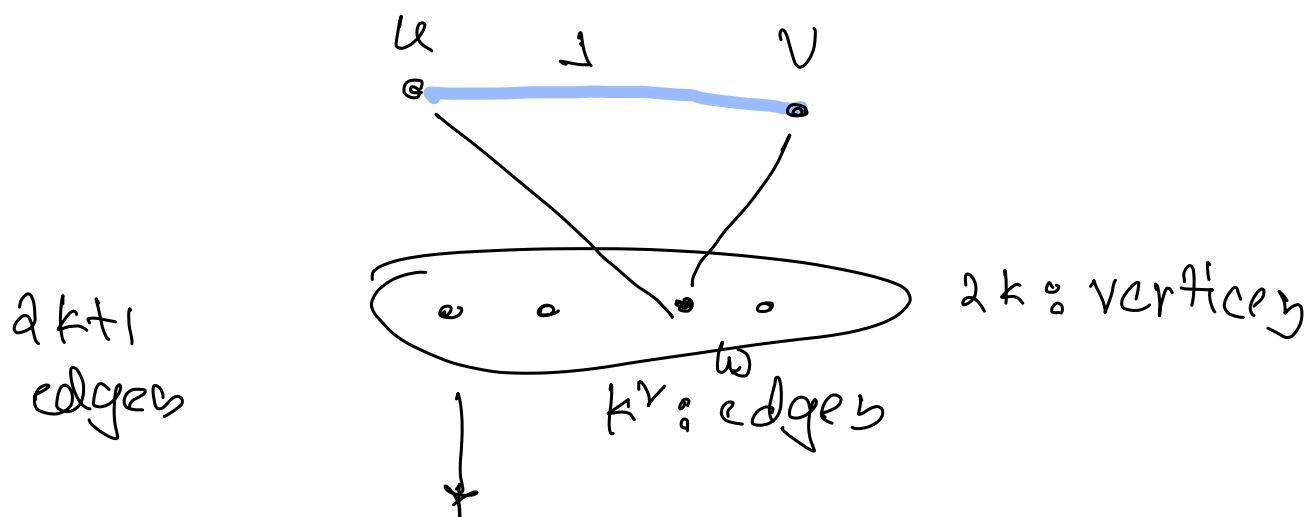
 $2k$ vertices
graph, H

there can be at max k^2 edges without
forming triangles

$$\text{remaining} = (k+1)^2 + 1 - k^2 = 2k + 2 = 2k + 1 + 1$$

I will use 1 to establish connection between, u and v

$$\text{remaining} = 2k+1$$



there are k no of pairs here,
connect one of each pair with an other other with v

I can at max form $2k$ connections
 $2k$ (edges)

by pigeonhole principle,

there is at least one vertex (say w)
which is connected to both u and v

hence, for $2k+2$ vertices with k^2+1 edges
we have a triangle,

(k_{k+1} statement)

conclusion: s_n is true for, $\forall n \in \mathbb{N}$